

Chapter # 01

Systems of Linear Equations

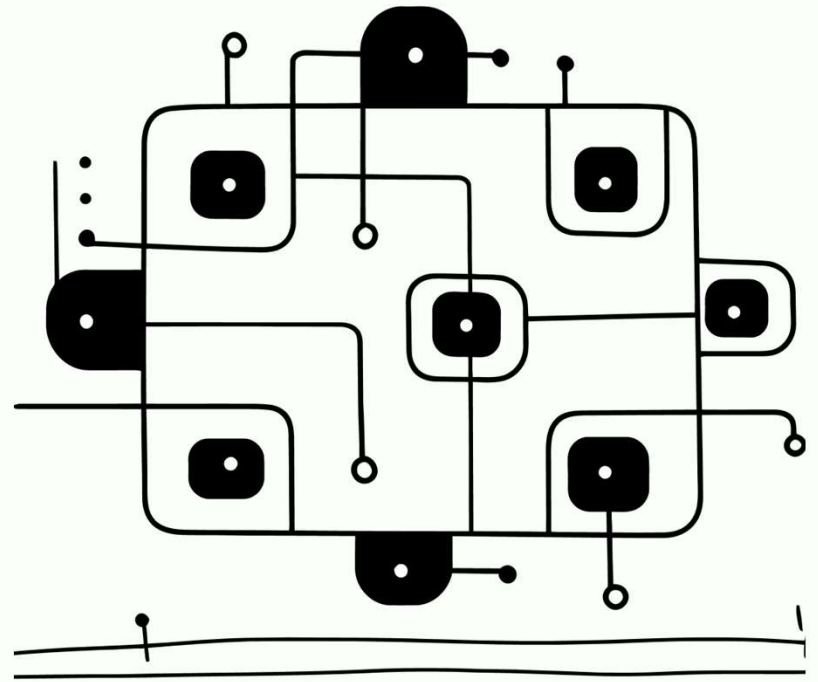
From equations to algorithms and real-world models

Course: Linear Algebra

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One mathematical language, many real problems

How can the same idea:



Balance combustion

Balance methane combustion reactions using stoichiometric constraints.



Locate a GPS receiver

Triangulate position from simultaneous distance equations.



Control traffic or irrigation flow

Model flow conservation at every junction as a linear constraint.



Fit a curve through data

Determine polynomial coefficients from data-point conditions.



Determine electrical current

Apply Kirchhoff's laws to find currents in a circuit network.

Each problem contains several unknown quantities linked by simultaneous linear constraints. The shared language is a **system of linear equations**.

Quick hook: If four unknowns must satisfy four independent clues, what outcomes are possible — one answer, many answers, or none?

LEARNING OUTCOMES

By the end of this chapter, you will be able to:

- 1 Classify equations**
Classify equations as linear or nonlinear.
- 2 Describe solution sets**
Describe solution sets parametrically and distinguish unique, infinitely many, or no solutions.
- 3 Form matrices**
Form coefficient and augmented matrices and identify matrix size, REF, and RREF.
- 4 Solve systems**
Solve systems of up to four variables using back-substitution, Gaussian elimination, or Gauss–Jordan elimination and verify the result.
- 5 Homogeneous systems**
Solve homogeneous systems and apply Theorem 1.1 correctly.
- 6 Build and interpret models**
Build and interpret systems for polynomial curve fitting and flow/electrical networks.

PRE-ASSESSMENT

What do you already know?

 **Instructions:** Work alone for 3 minutes; then compare with a partner.

1

Linear or not?

Which is linear: $3x + 2y = 7$,
 $xy + z = 2$, or $e^x - 2y = 4$?

2

Lines in a plane

How many solutions can two lines
in a plane have?

3

Solve the system

Solve: $x - 2y = 5, y = -2$.

4

Matrix dimensions

In a 3×4 matrix, how many rows and
columns are there?

5

Pre-assessment Solution

1 $3x+2y=7$ is linear

The other two contain a product of variables or a variable in an exponential function — both disqualify linearity.

2 Three possible outcomes

Two lines can intersect once, coincide and share infinitely many points, or be parallel and share none.

3 Back-substitution gives $x=1, y=-2$

Substitute $y=-2$ into $x-2y=5$: $x=5+2(-2)=1$.

4 Three rows and four columns

The notation $m \times n$ always means m rows by n columns.

Introduction to Systems of Linear Equations

In this section you will:

→ **Recognize a linear equation**

Identify a linear equation in n variables.

→ **Represent solution sets**

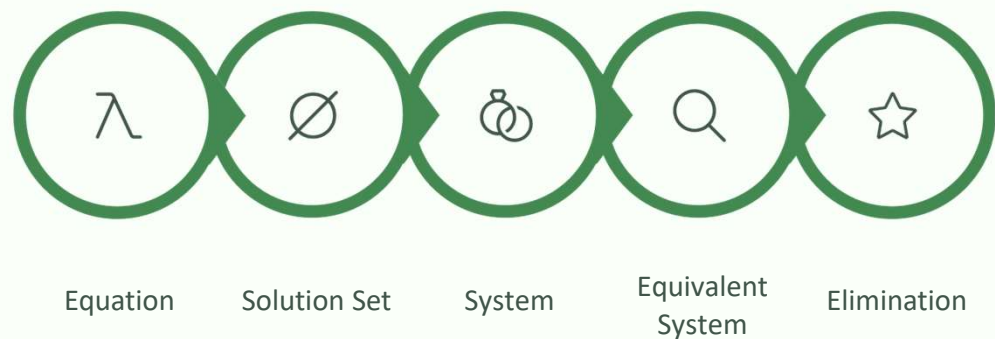
Represent a solution set with parameters.

→ **Classify consistency**

Decide whether a system is consistent or inconsistent.

→ **Solve by elimination**

Solve systems by back-substitution and Gaussian elimination.



DEFINITION

A linear equation has variables only to the first power

A linear equation in n variables has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where the coefficients a_i and constant b are real numbers.

Note that:

- a_i is coefficient of x_i
- **Leading coefficient and leading variable:** the first nonzero coefficient and its variable
- **Allowed:** real constants such as π , e^2 , $\sin \pi$
- **Not allowed:** products or roots of variables, powers other than 1, or variables inside trigonometric, exponential, or logarithmic functions

✓ Linear

$$3x + 2y = 7$$

All variables appear to the first power only.

✗ Not linear

$$xy + z = 2$$

Product of two variables — not allowed.

Example 1: Linear or Nonlinear?

✓ Linear

$$3x + 2y = 7$$

$$\frac{1}{2}x + y - \pi z = \sqrt{2}$$

$$(\sin \pi)x_1 - 4x_2 = e^2$$

$\sin \pi$ is a real constant — the variable x_1 is still to the first power.

✗ Not linear

$$xy + z = 2$$

Product of variables x and y .

$$e^x - 2y = 4$$

Variable x appears in an exponent.

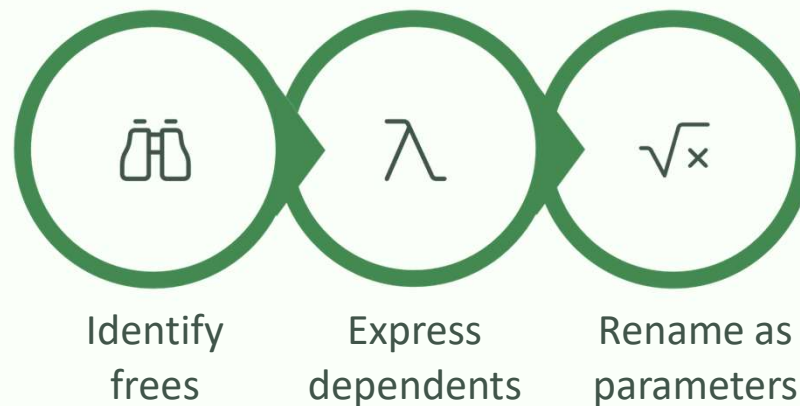
$$\sin x_1 + 2x_2 - 3x_3 = 0$$

Variable x_1 is inside a sine function.

Why is $(\sin \pi)x_1$ linear but $\sin x_1$ is nonlinear?

Solutions, solution sets, free variables, and parameters

A **solution** is an ordered sequence $(x_1, x_2, \dots, x_n)$ that makes an equation true after substitution. The **solution set** contains every solution.



When an equation has more variables than independent constraints, choose one or more **free variables**, write dependent variables in terms of them, and rename free variables with parameters such as s or t .

Example 2: One equation, infinitely many solutions

Solve $x_1 + 2x_2 = 4$.

Step-by-step solution

Solve for x_1 :

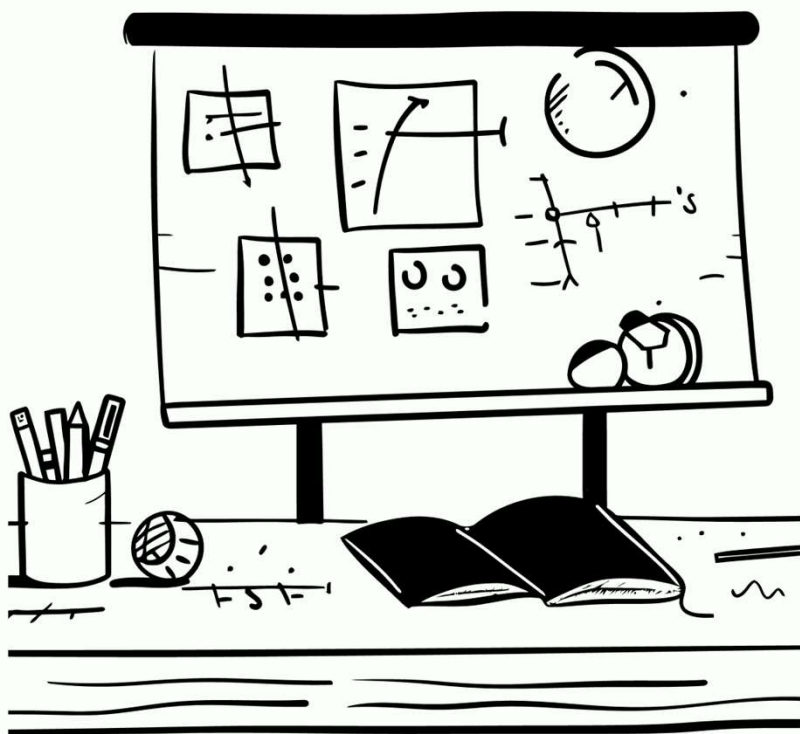
$$x_1 = 4 - 2x_2.$$

Let the free variable $x_2 = t$. Then

$$x_1 = 4 - 2t, \quad x_2 = t, \quad t \in \mathbb{R}.$$

Verification

- $t=1 \rightarrow (x_1, x_2) = (2, 1)$
- $t = 4 \rightarrow (x_1, x_2) = (-4, 4)$



Example 3: Two Free Variables

Solve $3x+2y-z=3$.

Choosing Free Variables

Choose y and z as free variables. Solving for x :

$$x = 1 - \frac{2}{3}y + \frac{1}{3}z$$

Let $y = s$ and $z = t$:

$$x = 1 - \frac{2}{3}s + \frac{1}{3}t,$$

Particular solutions include $(1,0,0)$ and $(1,1,2)$.

Active Check

Substitute both triples $(1,0,0)$ and $(1,1,2)$ into the original equation

$$3x+2y-z=3.$$

What geometric object is the complete solution set?

Systems of Linear Equations

Definition

A system of m linear equations in n variables has the form:

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i, \quad i = 1, \dots, m$$

a_{ij} means the coefficient of variable x_j in equation i .

Note that:

A **solution of the system** must satisfy **every equation simultaneously**.

Example: Consider the system:

$$3x_1 + 2x_2 = 3, \quad -x_1 + x_2 = 4$$

Testing Solutions

✓ $(-1, 3)$ is a solution

Satisfies **both** equations simultaneously.

✗ $(1, 0)$ is not

Satisfies only the **first** equation and fails the second.

Applied Mini-Task: Balance Methane Combustion

Model the reaction:



Conservation of Atoms

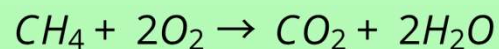
Setting up equations from atom counts:

Choose the smallest positive integers satisfying all three equations.



Pair Response 

$(x_1, x_2, x_3, x_4) = (1, 2, 1, 2)$, so:



Carbon

$$x_1 = x_3$$

Hydrogen

$$4x_1 = 2x_4$$

Oxygen

$$2x_2 = 2x_3 + x_4$$

Consistent or Inconsistent? Predict from the Graphs

Graph or mentally compare the following systems. A system is **consistent** when it has at least one solution and **inconsistent** when it has no solution.

System A

$$\begin{aligned} 3x - y &= 1 \\ 2x - y &= 0 \end{aligned}$$

System B

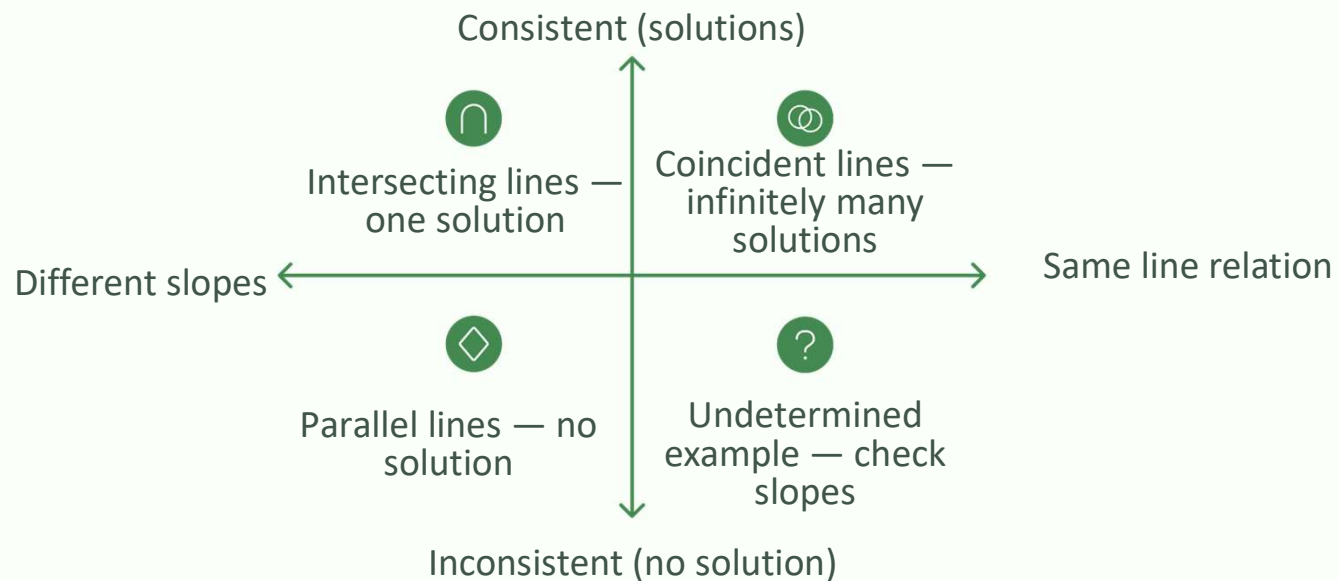
$$\begin{aligned} 3x - y &= 1 \\ 3x - y &= 0 \end{aligned}$$

System C

$$\begin{aligned} 3x - y &= 1 \\ 6x - 2y &= 2 \end{aligned}$$

Predict–Plot–Explain 🎨

Match A, B, and C to **intersecting**, **parallel**, and **coincident** lines before revealing the graphs. Discuss with a partner first!



Example 4: The Three Possible Solution Types

For any linear system, precisely one is true: **one solution, infinitely many solutions, or no solution.**

A linear system can never have exactly two isolated solutions.

1

Unique Solution

$$x + y = 3, \quad x - y = -1$$

→ unique solution (1,2);
two **intersecting** lines.

2

Infinitely Many Solutions

$$x + y = 3, \quad 2x + 2y = 6$$

→ infinitely many solutions;
the lines **coincide**.

$$x = 3 - t, \quad y = t$$

3

No Solution

$$x + y = 3, \quad x + y = 1$$

→ no solution;
the lines are **parallel**.